HEALTHIER, LONGER, BETTER LIVES

Can OCR Rules Replace Azure OpenAI?

Medical receipt extraction benchmark

9 receipts · 6 fields · rules vs AI extraction

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The Answer in Brief

Rules are enough for simple, clearly labelled checks, not as a general replacement for AOAI.

Not as a general replacement.

Rules work for simple, clearly labelled checks.

AOAI reduces the case-by-case logic we must write, and could unlock richer medical data.

88.2% vs 51.0–70.6%	Even with Chinese labels added (retrospective), rules trailed AOAI. The English-only first rules: 37.3–41.2%.
96.1% (development)	Clearer prompt instructions fixed four more fields. Not yet independently validated.
Less logic to maintain	Rules must anticipate every label, language and layout. AOAI works from plain-language instructions.

Small benchmark: 9 receipts. Development results are not independent validation.

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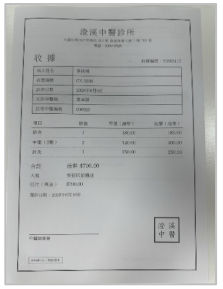
What We Compared

Three ways to turn a receipt photo into six fields. Only the extraction step differs between rules and AI.

	1. Read the text (OCR)		2. Find the fields		
PaddleOCR Light + rules	**PaddleOCR Light** open source, own compute	→	**Hand-written rules**	→	**6 fields**
Azure DI + rules	**Azure Document Intelligence** cloud OCR	→	**Hand-written rules**	→	**6 fields**
Azure DI + AOAI	**Azure Document Intelligence** cloud OCR	→	**Azure OpenAI instructions**	→	**6 fields**


English


Bilingual


Chinese

9 receipts: 7 printed test receipts (English, bilingual, Chinese layouts) and 2 from the first benchmark round.

51 fields scored: doctor, registration no., patient, receipt no., service date, total.

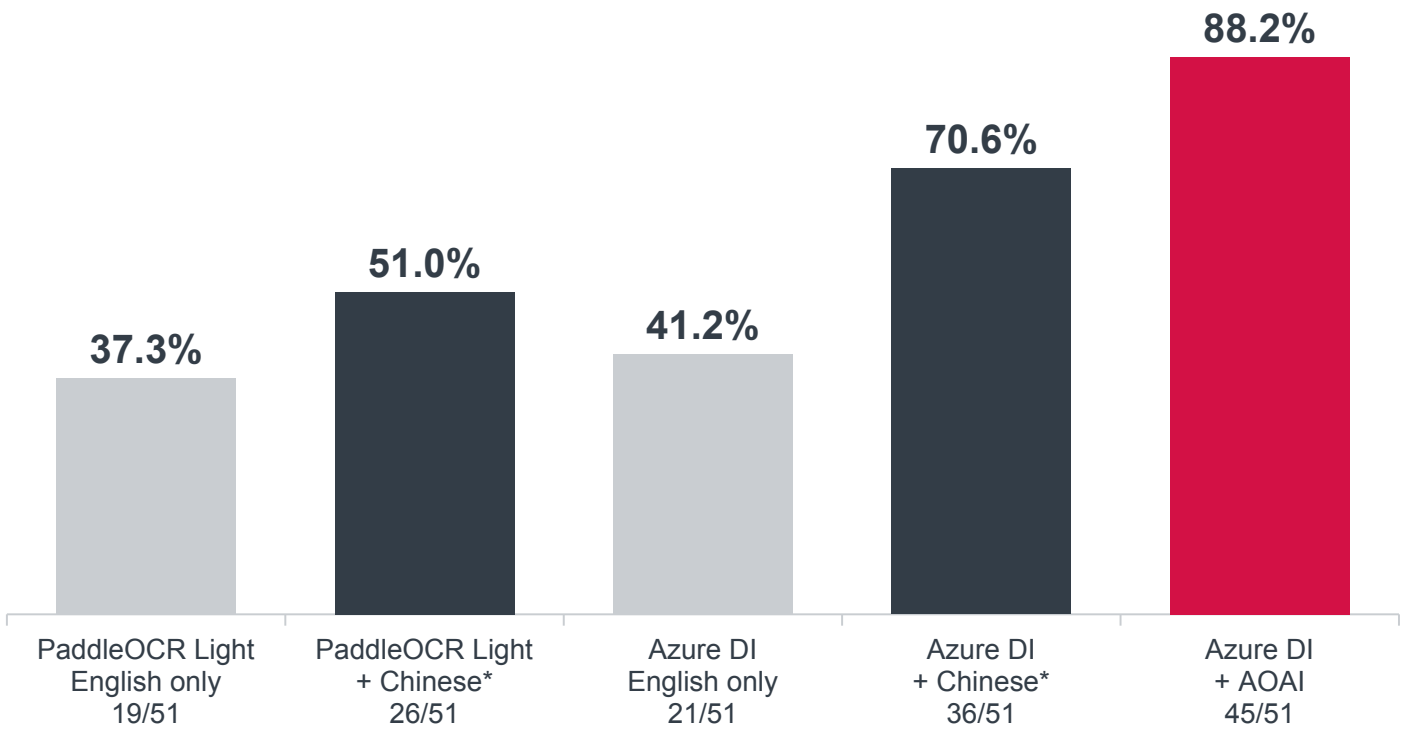
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Initial Results and a Fairer Rules Baseline

MEASURED · BEFORE TUNING

Even with Chinese labels added, rules trailed AOAI: 51.0–70.6% vs 88.2%.



Why the first rules scored low

- English labels only: no Chinese or bilingual labels.
- The OCR read 蔡嘉穎 ; the rules returned the label 病人 .

With Chinese labels (retrospective*)

- Common Chinese labels and basic bilingual handling: +15 fields on Azure DI, none broken.
- The rest would need new label types, layout logic and OCR-tolerant matching.

** Retrospective: Chinese labels added after these receipts were examined (rules fixed before scoring); not an independent test. Accuracy = correct fields out of 51 on 9 receipts.*

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Improving AOAI: Clearer Instructions

MEASURED · DEVELOPMENT SET

Prompt v2 fixed four fields: 88.2% → 96.1%. A development result, not independent validation.

Field fixed	Before	Instruction added	After
Registration GP receipt	MBBS (HK) · Reg. No. M17642	*"the identifier only, without its label"*	M17642
Doctor Dental receipt	Dr. Ho Ching Yee, BDS (HK)	*"Leave out qualifications and post-nominals"*	Dr. Ho Ching Yee
Doctor Pharmacy receipt	Dr Jessica Sterenborg Picton Medical Centre	*"clinic, hospital or practice names"*	Dr Jessica Sterenborg
Total Vet receipt	HKD 1361.50	*"after any discounts, rounding or other adjustments"*	HKD 1361.00

AOAI by field	Original	Prompt v2
Doctor	67% 6/9	**89%** 8/9
Registration*	83% 5/6	**100%** 6/6
Patient	100% 9/9	100% 9/9
Receipt no.	100% 9/9	100% 9/9
Service date	89% 8/9	89% 8/9
Total	89% 8/9	**100%** 9/9

*The gain came from the prompt wording; code checks, OCR layout and attaching the photo changed no result. * Registration is printed on 6 of 9 receipts.*

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The Key Difference: Maintenance

MEASURED · DEVELOPMENT SET

Rules need us to anticipate every case in code. AOAI can often cope through plain-language instructions.

Manual extraction rules

A new label, language or layout appears → A developer anticipates it and writes rule code → Retest every receipt; a change can break a working one

Seen in this benchmark

- 69 → 225 label phrases written by hand
- 1 correct field broken by a rule change

AOAI extraction

New wording appears → Often handled by the existing instructions → If not: refine the instructions in plain language, rerun

Seen in this benchmark

- One prompt revision: 4 fields fixed, 0 broken

Not guaranteed: AOAI has not been tested on enough unseen layouts. It still makes mistakes and needs testing and monitoring.

Where rules were enough: a yes/no check for a diagnosis comment under clear labels. Rules matched AOAI on 13/13 receipts (development result).

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Advantages and Disadvantages

Rules win on cost, auditability and predictability; AOAI on accuracy, maintenance and flexibility.

	OCR + rules	Azure DI + AOAI	Evidence
Cost	+ PaddleOCR: own compute, no API fee (Azure DI: page fee)	− Azure DI page fee + ~602 input / 263 output tokens per receipt	*Prices to confirm*
Speed	PaddleOCR 13.9 s (CPU) · Azure DI 3.8 s	6.2 s per receipt	Measured
Accuracy	− 37.3–41.2% English only; 51.0–70.6% with Chinese labels (retrospective)	+ 88.2% first; 96.1% with prompt v2 (development)	Measured
Maintenance	− New labels, languages and layouts need code and a full retest	+ Many changes made in plain-language instructions; still retest	*Measured / expected*
Flexibility	− Finds only what it was written for	+ New fields can be requested in the prompt (not yet benchmarked)	*Expected*
Auditability	+ Every result traceable to a rule	− Harder to explain; keep quoted evidence and logs	*Expected*
Reliability	+ Same input, same output	− Can return a confident wrong value (e.g. a sub total as the total); model updates need retesting	*Measured / expected*

Green = stronger side. First versions = before tuning; the first rules had no Chinese labels. Expected = not measured here.

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Beyond Six Fields: What AOAI Could Unlock

POTENTIAL · NOT YET TESTED

Our receipts already carry medical details. AOAI could turn them into data for analytics.

Type	Text already in our OCR	Receipt
Medication	Omeprazole 20mg Capsules	Pharmacy
Medication	Metronidazole 50mg Tablet	Vet invoice
Test	Chest X-ray	Hospital
Treatment	針灸 (acupuncture)	Chinese medicine
Treatment	Scaling and polishing	Dental
Symptom	diarrhoea	Vet invoice
Diagnosis	Colitis	Vet invoice

Which could enable

- Claims analytics by diagnosis, treatment and medication
- Cost benchmarks per treatment and provider
- Medication and treatment patterns over time
- Consistency checks: does the treatment fit the diagnosis and amount?
- Faster triage and fewer manual look-ups for reviewers

Potential, not yet benchmarked. Text shown is OCR input, not AOAI output. A pilot needs ground truth and a privacy review.

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Limits of This Evidence

The direction is clear, but the numbers are not yet proof.

1

Small dataset

9 receipts, 51 fields. 7 are printed test receipts.

2

First rules lacked Chinese

With Chinese labels added, rules reached 51.0–70.6% (retrospective, not an independent test).

3

Tuned on the same receipts

Prompt v2's 96.1% is a development result. Tuned rule scores are not shown.

4

Little independent validation

Only 4 unseen handwritten receipts. On all 4, Azure DI misread the year 2026 as 2020.

5

Costs not confirmed

Token counts are measured; Document Intelligence and gpt-5-mini prices are not.

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Recommendation

Use rules for simple, labelled checks. Use AOAI where receipts vary or richer data is needed.

Simple OCR rules are sufficient when…

- Layouts are few, known and stable
- Fields have clear printed labels
- The question is simple, e.g. is a diagnosis comment present?
- No API fee and fully traceable output are required

AOAI adds real value when…

- Providers, layouts and languages vary (Chinese, bilingual)
- Fields lack consistent labels: names with titles, totals vs subtotals
- We want less scenario-specific code to maintain
- Richer medical data is wanted (potential, not yet benchmarked)

1. Freeze

the rules and AOAI prompt v2

2. Validate

30+ unseen, real, anonymised receipts; count rule changes

3. Pilot extra fields

medications, treatments, symptoms, diagnoses with ground truth

4. Confirm cost

Document Intelligence pages and gpt-5-mini tokens

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